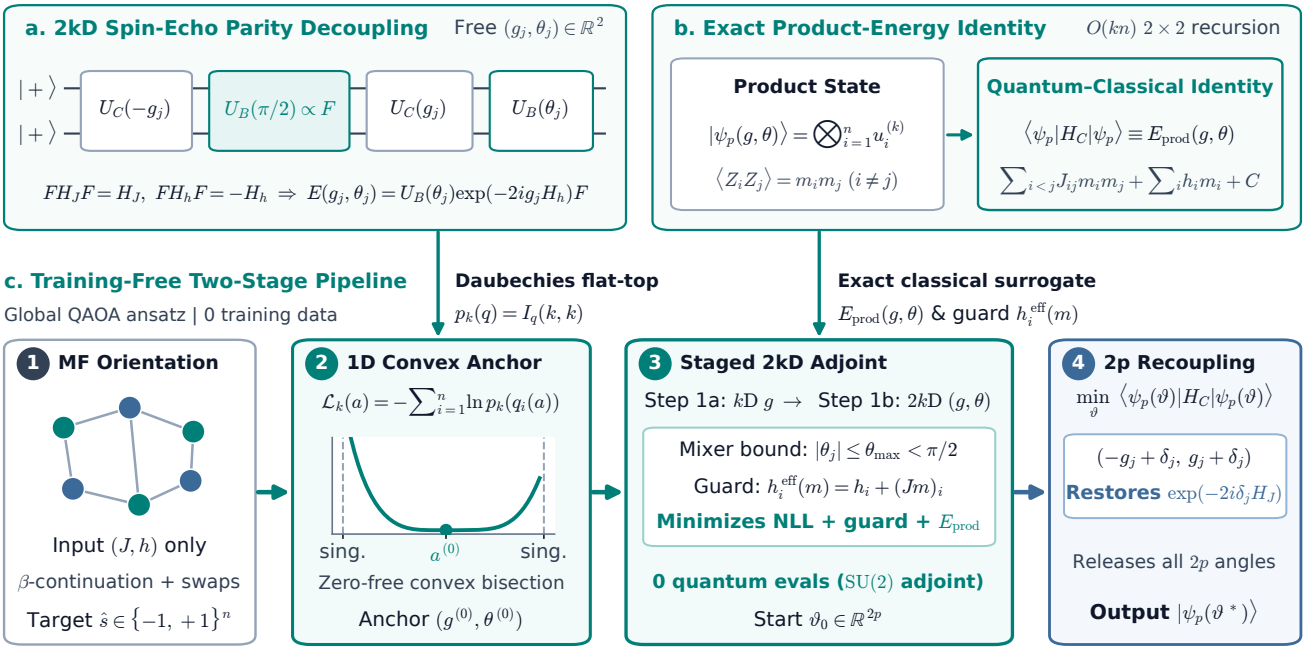

a. 2kD Spin-Echo Parity Decoupling
Free $(g_j, \theta_j) \in \mathbb{R}^2$
$|+\rangle$
$|+\rangle$
$U_C(-g_j)$
$U_B(\pi/2) \propto F$
$U_C(g_j)$
$U_B(\theta_j)$
$FH_JF = H_J,\ FH_hF = -H_h \Rightarrow E(g_j, \theta_j) = U_B(\theta_j)\exp(-2ig_jH_h)F$
b. Exact Product-Energy Identity
$O(kn)$ 2×2 recursion
Product State
$|\psi_p(g,\theta)\rangle = \bigotimes_{i=1}^{n} u_i^{(k)}$
$\langle Z_iZ_j \rangle = m_im_j\ (i \neq j)$
Quantum-Classical Identity
$\langle \psi_p | H_C | \psi_p \rangle \equiv E_{\text{prod}}(g, \theta)$
$\sum_{i<j} J_{ij}m_im_j + \sum_i h_im_i + C$
c. Training-Free Two-Stage Pipeline
Global QAOA ansatz | 0 training data
Daubechies flat-top
$p_k(q) = I_q(k, k)$
Exact classical surrogate
$E_{\text{prod}}(g, \theta)$ & guard $h_i^{\text{eff}}(m)$
1 MF Orientation
Input (J, h) only
β-continuation + swaps
Target $\hat{s} \in \{-1, +1\}^n$
2 1D Convex Anchor
$\mathcal{L}_k(a) = -\sum_{i=1}^{n} \ln p_k(q_i(a))$
sing.
$a^{(0)}$
sing.
Zero-free convex bisection
Anchor $(g^{(0)}, \theta^{(0)})$
3 Staged 2kD Adjoint
Step 1a: kD g → Step 1b: $2k$D (g, θ)
Mixer bound: $|\theta_j| \leq \theta_{\max} < \pi/2$
Guard: $h_i^{\text{eff}}(m) = h_i + (Jm)_i$
Minimizes NLL + guard + E_{prod}
0 quantum evals (SU(2) adjoint)
Start $\vartheta_0 \in \mathbb{R}^{2p}$
4 2p Recoupling
$\min_{\vartheta} \langle \psi_p(\vartheta) | H_C | \psi_p(\vartheta) \rangle$
$(-g_j + \delta_j,\ g_j + \delta_j)$
Restores $\exp(-2i\delta_jH_J)$
Releases all $2p$ angles
Output $|\psi_p(\vartheta^*)\rangle$